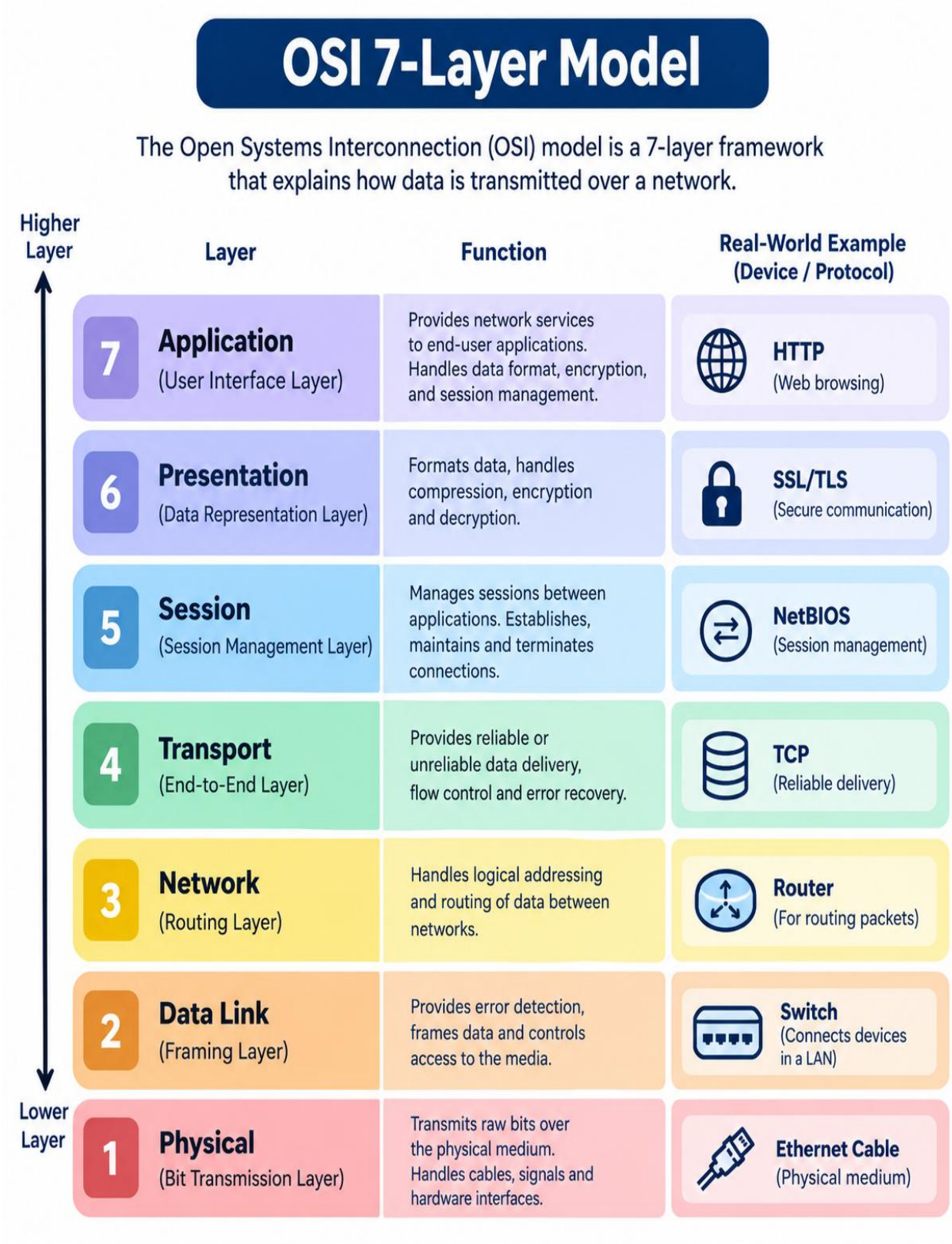
OSI layer

Q-1 Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

Ans:


OSI 7-Layer Model
The Open Systems Interconnection (OSI) model is a 7-layer framework that explains how data is transmitted over a network.
Higher Layer
Lower Layer
Layer
Function
Real-World Example (Device / Protocol)
7
Application
(User Interface Layer)
Provides network services to end-user applications. Handles data format, encryption, and session management.
HTTP
(Web browsing)
6
Presentation
(Data Representation Layer)
Formats data, handles compression, encryption and decryption.
SSL/TLS
(Secure communication)
5
Session
(Session Management Layer)
Manages sessions between applications. Establishes, maintains and terminates connections.
NetBIOS
(Session management)
4
Transport
(End-to-End Layer)
Provides reliable or unreliable data delivery, flow control and error recovery.
TCP
(Reliable delivery)
3
Network
(Routing Layer)
Handles logical addressing and routing of data between networks.
Router
(For routing packets)
2
Data Link
(Framing Layer)
Provides error detection, frames data and controls access to the media.
Switch
(Connects devices in a LAN)
1
Physical
(Bit Transmission Layer)
Transmits raw bits over the physical medium. Handles cables, signals and hardware interfaces.
Ethernet Cable
(Physical medium)

Q- 2 Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.

Ans:

a) Encrypting data before sending: Presentation Layer (Layer 6)

b) Routing packets across networks: Network Layer (Layer 3)

c) Converting data to electrical signals: Physical Layer (Layer 1)

d) Opening a web page in your browser: Application Layer (Layer 7)

e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)

Q-3 Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

Ans: Example: WhatsApp

When I send a message to a friend on WhatsApp, different OSI layers work together to deliver the message.

1. Application Layer (Layer 7): WhatsApp allows me to type and send messages to my friend.

2. Transport Layer (Layer 4): Protocols such as TCP or UDP help transport data between devices, depending on the communication method used.

3. Network Layer (Layer 3): IP addresses help route data packets from my device to the destination through the network.

4. Data Link Layer (Layer 2): It transfers data across the local network, such as Wi-Fi, and helps detect transmission errors.

5. Physical Layer (Layer 1): It transmits data as electrical, radio, or optical signals through cables or wireless connections.

Q-4 Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Ans: OSI Layers Causing the Problem

If I can successfully ping a website's IP address but cannot access the website, the problem may be in these OSI layers:

1. Application Layer (Layer 7): The website or web server may have an application-level problem.
2. Transport Layer (Layer 4): The TCP connection or web service ports, such as HTTP (80) or HTTPS (443), may be blocked or unavailable.
3. Presentation Layer (Layer 6): There may be an SSL/TLS certificate or secure connection problem.

Reasoning: Successful pinging shows that IP connectivity is working to some extent, but it does not guarantee that the website's web service is available. The issue could also involve DNS if the website is accessed by its domain name.